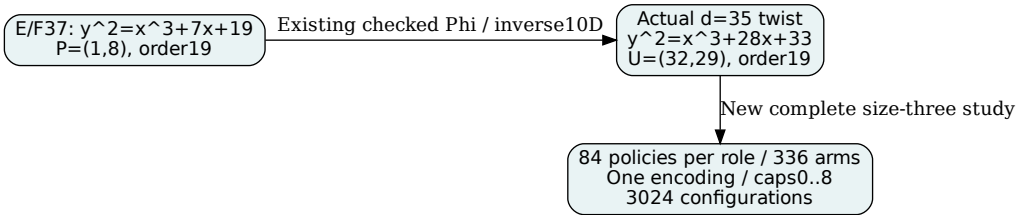
Complete size-three factor bases over $\mathbf{F}_{37}$

2026-10-30 PR2381 fixed public synthetic inputs

The existing actual route is $E : y^2 = x^3+7x+19$ to its nonsquare $d = 35$ monic twist $T : y^2 = x^3+28x+33$. The checked coordinate maps give $P = (1, 8) \mapsto U = (32, 29)$ and inverse $[10]D$ on the cyclic subgroups of order 19. These are exact unregistered model tuples, not invented registry aliases.



The new cohort contains all size-three subsets of nine unsigned subgroup abscissae: 84 canonical policies per curve, plus every matched and pullback control. All 336 arms run caps 0–8 under one original encoding. The 3024 configurations make 54432 separate target attempts on 18 repeated source points. These are not 54432 distinct DLP challenges.

Encoding control. S_3 has degree at most two in each coordinate, although its total degree is four. Reducing it by a cubic product-membership polynomial in either coordinate leaves it unchanged. The producer checks the coefficient bounds; independent Rust performs monic division and checks equality on every query. A second identical encoding is not counted as an extra arm.

New Lean check. The unmodified scoped gate passes six public named statements / seven audited declarations, zero failing or extra trusted/native axioms. Over a nontrivial commutative ring it proves cubic monicity/degree, the univariate degree bound for the explicit S_3, remainder identity, both coordinate versions and binding to the native value formula. Other coordinates are parameters. This is an input-equality theorem, not a solving-degree, intrinsic-regularity or DLP-hardness theorem. The actual source-DLP reduction uses the earlier F37 19-named/22-audited receipt within its cyclic domain; that receipt is not a newly combined audit.

Independent native outcome	Count
Configurations	3024
Verified original-source recovery runs	1944
Blocked public-log setups	2916
Capped diagnostic query records	52100
Policy/target/requested-cap pair records	10287648
Strict higher-rank/lower-cap recovery-admitted records	0
Such records beating strongest source policy	0
Changed controls rejected 101/no receipt	10
Shared backend tests passed	22

Recorded outcomes and limits

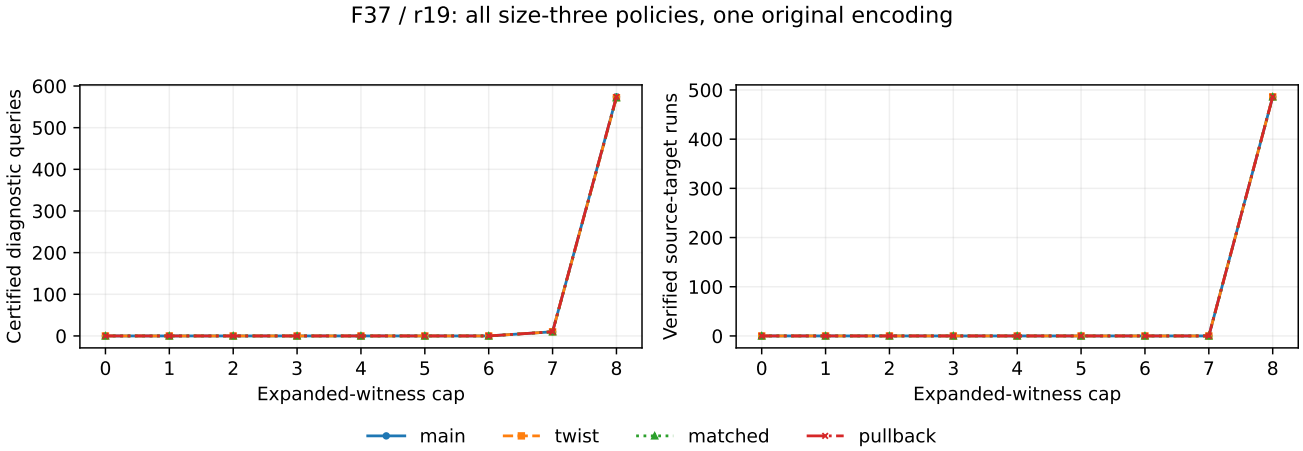


Figure. Native-emitted counts only: 84 policies and 18 source points per role and cap. Curves may overlap. These are exact finite counts; no timing, uncertainty estimate or cross-size performance ratio is inferred.

Public setup derives logs only from admitted rows until coefficient rank three, then checks each base point. Incomplete rank blocks every target. Recovery receives actual unknown Q, uses fresh shifted solves, retains failed and identity shifts and all ordered sign lifts, and verifies the scalar on original E/P. The finite recovery count repeats the same 18 source points.

The new solver uses S_3 and two cubic memberships. Matrix and expanded original-input verification-witness degrees must fit the rung before shifts are constructed. A complete unit or bounded Buchberger certificate precedes root extraction; unknown caps admit no roots or rows. A smaller already-found witness is retained only for an identical polynomial within one solve. Independent arithmetic reconstructs every case, witness, row, rank, setup log, shift and source scalar. A separate degree-free point reference never controls candidate stopping. Diagnostic output binding follows fresh solves and is unpriced. Setup, field/group operations, allocation and verification lack a complete price.

Both full native paths pass. False inverse plus nine genuinely changed/rehashed case controls reject 101 without receipts. The same 22 backend tests pass. All four native panels record zero recoveries at caps 0–7 and 486 per role at cap 8. Hence no admitted target has a higher source comparison cap available inside 0–8. The zero joint count is boundary-limited, not a proof of universal no-advantage. The lower cap is a requested budget, not automatically an actual solving degree. Stationary useful yield, intrinsic regularity and calibrated complete-cost gains remain open.

Preservation, integration and next lead

All cases and altered payloads remain source-bound. Raw sources, commands, exits, failures and proof receipts are archived. Full control trees reconstruct from canonical case files plus exact overlays/removals; their bytes match the originals. Canonical protocol and export hashes bind the new full scope.

Native stages and launcher explicitly record zero; an outer PTY result reported one despite the explicit file. All stage/rejection receipts and ordinary metadata checks resolve the distinction without a repeated study. Lean attempt 01 fails on abstract-polynomial computability and ambiguous binomial notation. Attempt 02 fails on multiplication order and an unmatched section end. Same statements fixed by a noncomputable section, disambiguated syntax, ring proof and explicit section closure pass on attempt 03. Failed sources/logs remain; no sorry or trust bypass is accepted.

Owner merge review. A new normal main merge leaves every frozen study path unchanged. Active registry records increase 20787 to 20809 by 22 additions, with no old record removed or shared value changed; the 1921 lost study rows remain absent. The PR native-audit failure again finds 1921 proposed rows and 3842 aliases. Its actual checkout is a synthetic merge 26fb0d38…, not the literal PR head; their registry hashes are distinguished. Existing 42 metadata tests and seven separate replay jobs pass. Required checks are assessed on each resulting head; older passes do not replace them.

The owner re-enabled auto-merge. A dedicated non-merge GraphQL mutation restored the user's keep-open state. A CLI spelling was rejected by automatic approval review before execution. The local fast-forward initially ran out of disk; only reproducible build cache was removed. Every partial file was verified against the owner tree before checkout repair. No user changes or evidence were lost. Root/catalog/naming/coordination/review holds remain.

Next test, not executed. Preregister fresh caps 9–12 across all 336 arms. Retain 0–8 as fixed historical evidence, explicitly separating old and new executions. This creates comparisons above the observed cap 8 admission boundary; it must not silently change this frozen contract. Preserve all blocked/failed setups and the full 84-policy source frontier. A symmetric-coordinate membership encoding with distinct-root and diagonal branches is a secondary hypothesis: equivalence, lifting and degree accounting need proof before any such run. Neither extension nor new encoding is implemented or executed here.

Scientific window 15:15–16:15 UTC: two existing local workers, 600-second command caps, no paid resources or arbitrary result upload. Keep the same PR open, non-draft, with auto-merge off. The original stronger goal remains open. Full proof map, exact hypotheses, native receipts, source links, restoration commands and canonical flow are in the adjacent README/validation.